

Project Development Phase

Model Performance Test

Project Name	AI-Enhanced Intrusion Detection System
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Model Performance Testing:

Project team has filled the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Model Summary	Random Forest Classifier trained on CICIDS2017 Dataset. Optimized to 4 key features (Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets). Class imbalance handled with SMOTE.	
2.	Accuracy	Training Accuracy – ~99.2% Validation Accuracy – ~98.9%	
3.	Confidence Score (Only Yolo Projects)	Class Detected – N/A (Non-YOLO project) Confidence Score – N/A	

Notes:

- The model uses Random Forest with 4 features: Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets.
- SMOTE was applied to handle class imbalances in the CICIDS2017 dataset.
- Verdict classes: BENIGN, Web Attack – Brute Force, Web Attack – SQL Injection, Web Attack – XSS.